RAJALAKSHMI ENGINEERING COLLEGE

Shalini G V 251001244-2025-IT S2

Started on	Thursday, 13 August 2026, 8:53 PM
State	Finished
Completed on	Thursday, 13 August 2026, 8:56 PM
Time taken	2 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int floorValue(int a[], int low, int high, int x)
{
    if (low > high)
        return -1;
    int mid = (low + high) / 2;
    if (a[mid] == x)
        return a[mid];
    if (a[mid] > x)
        return floorValue(a, low, mid - 1, x);
    int result = floorValue(a, mid + 1, high, x);
    if (result == -1)
        return a[mid];
    return result;
}
int main()
{
    int n, i, x, a[100];
    scanf("%d", &n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    printf("%d", floorValue(a, 0, n - 1, x));
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔

	Input	Expected	Got	
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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